USN [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] **[22BS2MA03]**

SECOND SEMESTER B.Tech EXAMINATION, JULY - 2023

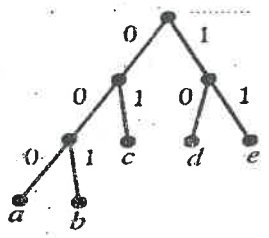
DISCRETE MATHEMATICS AND GRAPH THEORY

Time: 3 Hours **Maximum Marks: 100**

Instructions: Missing data may be suitably assumed.

Answer all questions PART – A **5 X 2=10**

1.	**a.**	There are 20 married couple in a party. Find the number of ways of choosing one woman and one man from the party such that the two are not married to each other.	L-2	CO3	**02**
	b.	Find the sequence generated by the function $2x^2(1-x)^{-1}$.	L-2	CO4	**02**
	c.	Find General solution of homogeneous part of the recurrence relation $a_n - a_{n-1} - 2a_{n-2} = 4$	L-3	CO4	**02**
	d.	Define spanning subgraphs and induced subgraphs.	L-1	CO1	**02**
	e.	Obtain the prefix code represented by the following labeled complete binary sequence.	L-2	CO1	**02**



PART - B **6 X 5=30**

2.	**a.**	From the set of all permutation of n distinct objects, one permutation is chosen at random. What is the probability that it is not a derangement?	L-3	CO3	**05**
		OR			
	b.	Find the rook polynomial for a 3× 3 board	L-3	CO3	**05**
3.	**a.**	Prove that the number of partitions of a positive integer n into distinct summands is equal to the number of partitions of n into odd summands.	L-3	CO4	**05**
		OR			
	b.	Determine the coefficient of x^{15} in $\frac{(1+x)^4}{(1-x)^4}$	L-3	CO4	**05**
4.	**a.**	Suppose that there are $n \geq 2$ persons at a party and that each of these persons shakes hands (exactly once) with all of the other persons present. Using a recurrence relation find the number of hand shakes.	L-3	CO4	**05**

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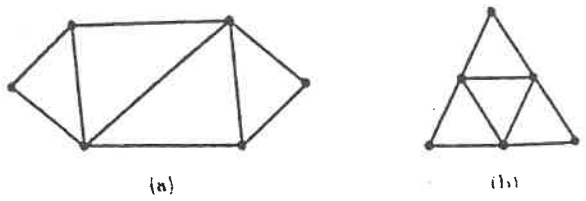
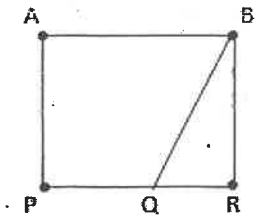
		OR			
	b.	Solve the recurrence relation $a_n = 7a_{n-1} - 10a_{n-2}$ with $a_0 = 2$ and $a_1 = 3$.	L-3	CO2	**05**
5.	**a.**	Determine whether following graphs are isomorphic or not  (a) (b)	L-2	CO5	**05**
		OR			
	b.	Find the chromatic polynomial for the cycle $C4$ of length 4	L-2	CO5	**05**
6.	**a.**	Define the following with suitable example. **i.** Tree **ii.** Leaf **iii.** Node	L-2	CO5	**05**
		OR			
	b.	Define Path. Consider the graph shown below. Find all paths from vertex A to vertex R. also, indicate their lengths. 	L-2	CO5	**05**
7.	**a.**	Show that if 30 dictionaries in a library contain a total of 61,327 pages, then one of the dictionaries must have at least 2045 pages.	L-3	CO2	**05**
		OR			
	b.	Determine the coefficient of x^{18} in the expression $(x + x^2 + x^3 + x^4 + x^5)(x^2 + x^3 + x^4 + \cdots)^5$	L-3	CO1	**05**
		PART - C	**6 X 10=60**		
8.	**a.**	Determine the number of positive integers n such that $1 \leq n \leq 100$ and n is not divisible by 2,3 or 5?	L-3	CO3	**10**

OR

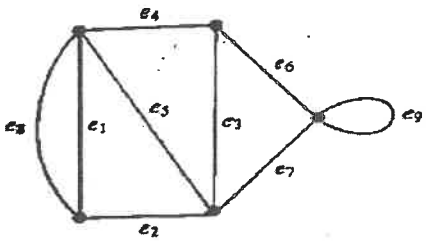
	b.	Five teachers T1, T2, T3, T4, T5 are to be made class teachers for five class C1, C2, C3, C4, C5, one teacher for each class. T1 and T2 do not wish to become the class teachers for C1 or C2, T3 and T4 for C4 or C5, and T5 for C3 or C4 or C5. In how many ways can the teachers be assigned the work (without displeasing any teacher)?	L-3	CO2	**10**
9.	**a.**	For each of the words given below, find the exponential generating function for the number of ways to arrange ns 11 letters selected from the word. **i.** MISSISSIPPI **ii.** ISOMORPHISM **iii.** MATHEMATICS **iv.** ENGINEERING	L-3	CO4	**10**

OR

	b.	Using generating function, find the number of partitions of n = 6. Also write all possible partitions of n=6.	L-3	CO4	**10**
10.	**a.**	Solve the recurrence relation $a_{n+2} - 10a_{n+1} + 21a_n = 3n^2 - 2,\ n \geq 0.$	L-3	CO4	**10**

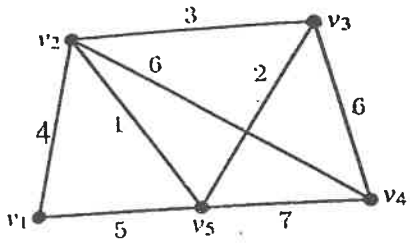
OR

	b.	Find the generating function for the recurrence relation $a_{n+1} - a_n = n^2,\quad n \geq 0,$ with $a_0 = 1$. Hence solve it.	L-3	CO4	**10**
11.	**a.**	Carry out the elementary reduction process for the following graph shown in below Figure and check the planarity.	L-2	CO5	**10**



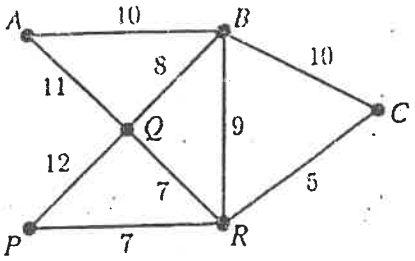
OR

	b.	Explain the following with the an example **i.** Graph **ii.** Simple Graph **iii.** Multigraph **iv.** Complete Graph **v.** Regular Graph.	L-2	CO5	**10**
12.	**a.**	Using Prim's algorithm, find a minimal spanning tree of the weighted graph.	L-2	CO5	**10**



OR

b.	**i.**	Using the merge sorting method sort the list {12, 11, 2, 3, 1, 5, 6, 7, 10, 9, 4, 15}	L-2	CO5	**10**
	ii.	If a tree T has 3 vertices of degree 2, 2 vertices of degree 3 and 2 vertices of degree 4. Find the number of pendant vertices in T.			
13.	**a.**	Explain the following with an example.	L-2	CO5	**10**
	i.	Walk			
	ii.	Trail			
	iii.	Path			
	iv.	Circuit			
	v.	Cycle			

OR

b.		Using Kruskal's Algorithm, find a minimal spanning tree for the given weighted graph.	L-2	CO5	**10**